

SIWA, Egypt

WMO: 624170

Lat: 29.200N

Lon: 25.483E

Elev: -15

StdP: 101.51

Time Zone: 2.00 (E02)

Period: 95-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	4.9	6.1	-4.8	2.5	21.4	-2.7	3.0	18.9	9.4	16.6	8.4	16.3	0.5	200	0.371

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	12.7	41.3	21.2	39.8	21.0	38.5	20.9	23.4	34.4	22.9	34.0	22.5	33.6	3.1	330

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
20.6	15.2	27.5	19.8	14.5	27.6	19.0	13.8	27.7	69.4	34.2	67.5	33.9	65.8	33.7	27.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
7.9	6.6	5.6	2.6	44.1	1.2	1.2	1.8	44.9	1.1	45.7	0.4	46.4	-0.4	47.3		
			0.7	24.6	1.3	1.2	-0.2	25.5	-1.0	26.1	-1.8	26.8	-2.7	27.7		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	23.1	13.4	14.9	18.8	22.8	26.8	30.1	31.4	31.5	29.2	24.5	18.8	14.8
	DBStd	6.97	2.05	2.70	3.25	3.23	3.09	2.38	1.73	1.51	2.00	2.60	2.55	2.01
	HDD10.0	1	1	0	0	0	0	0	0	0	0	0	0	0
	HDD18.3	421	155	102	31	2	0	0	0	0	0	22	110	
	CDD10.0	4796	105	138	274	383	519	603	664	667	577	449	263	150
	CDD18.3	2174	1	6	47	136	261	353	406	409	327	191	35	2
	CDH23.3	28645	18	84	522	1443	3233	4987	5998	6021	4215	1806	302	17
	CDH26.7	16361	3	25	199	649	1782	3057	3759	3717	2355	746	68	1
Wind	WSAvg	2.0	2.0	2.5	2.6	2.7	2.4	2.1	2.0	1.9	1.8	1.4	1.3	1.7
Precipitation	PrecAvg	7	1	1	1	1	0	0	0	0	0	1	1	1
	PrecMax	24	7	8	8	14	6	0	0	2	3	7	16	7
	PrecMin	0	0	0	0	0	0	0	0	0	0	0	0	0
	PrecStd	6	2	2	2	3	1	0	0	0	1	1	3	2
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	25.6	29.6	35.5	38.4	42.1	43.9	43.2	41.9	40.6	36.6	31.7	25.5
		MCWB	13.3	13.9	16.2	17.8	19.8	21.5	21.8	22.3	21.1	20.4	18.3	14.1
	2%	DB	22.2	25.7	31.4	35.2	38.8	40.9	40.9	40.1	38.7	34.0	28.5	22.9
		MCWB	12.5	13.1	15.6	16.8	19.3	20.7	21.5	21.8	21.2	19.9	17.4	14.6
	5%	DB	20.3	23.1	28.4	32.9	36.6	39.0	39.2	38.7	36.9	32.2	26.4	21.6
		MCWB	12.3	12.8	14.6	16.8	18.8	20.4	21.2	21.7	21.1	19.7	16.7	14.0
	10%	DB	19.1	21.0	25.9	30.5	34.6	37.2	37.8	37.5	35.5	30.7	24.8	20.4
		MCWB	11.8	12.3	14.0	16.0	18.4	20.2	21.1	21.5	20.8	19.3	16.0	13.6
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	14.7	15.4	18.1	19.4	21.5	22.9	23.8	24.2	23.7	22.3	19.7	16.9
		MCDB	21.6	25.3	28.9	34.3	36.8	37.4	35.3	35.2	34.6	31.6	28.1	22.3
	2%	WB	13.7	14.2	16.5	18.1	20.4	22.1	23.1	23.6	22.9	21.2	18.5	15.7
		MCDB	20.1	22.7	28.4	32.3	35.3	36.5	35.4	34.1	33.1	30.2	26.7	21.3
	5%	WB	12.9	13.4	15.3	17.2	19.6	21.5	22.6	23.1	22.3	20.6	17.5	14.7
		MCDB	19.3	21.3	26.5	30.3	33.9	35.6	34.7	33.5	32.4	29.6	25.0	20.6
	10%	WB	12.2	12.7	14.5	16.4	19.0	20.9	22.1	22.6	21.7	20.0	16.6	13.8
		MCDB	18.4	20.1	24.8	28.8	32.6	34.5	33.8	33.2	31.9	29.0	23.7	19.7
Mean Daily Temperature Range	5% DB	MDBR	11.9	11.9	12.8	13.5	14.0	14.1	13.6	12.7	12.5	11.8	12.2	11.6
		MCDBR	13.7	15.3	16.3	17.0	16.5	15.9	15.2	14.2	14.4	13.4	13.9	12.8
		MCWBR	7.1	6.4	5.9	5.7	5.2	4.4	4.0	4.0	4.3	4.9	6.3	6.9
	5% WB	MCDBR	12.4	12.5	14.8	15.0	14.7	14.6	13.9	12.5	12.4	11.9	12.7	11.9
		MCWBR	7.0	6.0	5.9	5.4	5.0	4.4	4.1	3.9	4.2	4.9	6.3	7.0
Clear-Sky Solar Irradiance	taub	0.395	0.450	0.499	0.568	0.573	0.504	0.467	0.470	0.493	0.454	0.404	0.393	
	taud	2.021	1.856	1.732	1.616	1.631	1.800	1.909	1.909	1.845	1.943	2.039	2.039	
	Ebn at Noon	815	799	788	750	748	797	826	821	784	786	799	793	
	Edh at Noon	145	187	226	263	260	218	195	193	199	169	142	137	
All-Sky Solar Radiation	RadAvg	3.84	4.70	5.93	6.87	7.50	8.01	7.88	7.31	6.43	5.26	4.15	3.56	
	RadStd	0.10	0.24	0.17	0.16	0.23	0.15	0.08	0.08	0.09	0.10	0.10	0.13	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB		1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (0 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page